

Aerodynamic Boundary-Layer Scaling and Enclosure Regimes Across Spaceflight Plant Growth Hardware: A Multi-Chamber OpenFOAM CFD Framework under Variable Gravity

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ABSTRACT

Plants cultivated in extraterrestrial habitats encounter a physical environment devoid of natural gravity-driven buoyancy ($Gr \rightarrow 0$), expanding unstirred fluid boundary layers around vegetative canopies and drastically elevating aerodynamic resistance ($r_a = \frac{1}{g_{bl}}$). Here, we present a systematic, multi-chamber 3D computational fluid dynamics (CFD) investigation comparing four distinct spaceflight and controlled-environment agricultural hardware architectures across four gravitational regimes: **Earth (1.0 g)**, **Mars (0.38 g)**, **Moon (0.166 g)**, and **Microgravity (0 g)**. Using an OpenFOAM v2606 finite-volume framework with conformal multi-solid analytic geometries, we model: (i) the compact **Microgreen Chamber** (2.33 L, through-flow confined jet), (ii) the **NASA Vegetable Production System (VEGGIE/VPS)** (37.6 L, top suction with passive cabin air induction), (iii) the **NASA Advanced Plant Habitat (APH)** (83.4 L, ducted closed-loop opposing cross-flow), and (iv) the **NASA Space Shuttle CHROMEX / Plant Growth Unit (PGU/PGC)** (49.57 L macro chassis, 0.866 L canisters with Brinkman-Darcy rooting foam). Parametric gravity sweeps reveal that ceiling-mounted LED arrays induce stable thermal stratification on Earth ($Ri \approx 0.14 - 1.55$), which suppresses vertical exchange; in microgravity, this stratification collapses, rendering purely forced convection ($Ri = 0$) superior in turbulent kinetic energy. In VEGGIE, low-fan microgravity operation leads to a critical 52.8% canopy stagnation volume ($g_{bl} = 0.219 \text{ mol m}^{-2}\text{s}^{-1}$), elevating fungal mold vulnerability. In CHROMEX sealed canisters, pure diffusion ($Pe < 1$) drives root-zone hypoxia ($O_2 < 5\%$) within 35 minutes, providing a biophysical basis for historical flight transcriptomic alcohol dehydrogenase (**ADH**) upregulation. Transient fan-stoppage tests reveal that on Earth, natural buoyancy maintains a basal conductance floor ($g_{bl} \approx 0.36 \text{ mol m}^{-2}\text{s}^{-1}$), whereas in microgravity, total aerodynamic collapse suffocates the canopy within 3.5–8.9 minutes. Conversely, APH maintains invariant $g_{bl} \approx 1.07 \text{ mol m}^{-2}\text{s}^{-1}$ across all gravities.

Introduction & Biophysical Foundations

Opportunities & Imperatives of Space Agriculture

As human space exploration transitions from low-Earth orbit sorties toward sustained surface outposts on the Moon (NASA Artemis Base Camp) and multi-year transits to Mars, biological life support systems become indispensable (Wheeler 2017). Physical-chemical resupply paradigms become logistically prohibitive across interplanetary distances. Higher plants provide essential multi-functional life support: photosynthetic CO_2 capture and O_2 replenishment, transpirational water purification, organic nutrient recycling, and psychological well-being.

Microgravity Fluid Mechanics & Buoyancy Cessation

Despite these compelling opportunities, cultivating crops in extraterrestrial environments confronts a fundamental physical impediment: the total cessation of gravity-driven natural convection (Kitaya et al. 2001, 2003; Porterfield 2002). On Earth (1.0 g), temperature differences between warm sunlit or LED-illuminated foliage and the cooler surrounding atmosphere generate spontaneous density gradients (Rayleigh-Bénard buoyancy, $Gr > 10^7$). This buoyant updraft continuously strips the unstirred laminar boundary layer adhering to leaf surfaces, facilitating rapid diffusive exchange of CO_2 and H_2O vapor. In microgravity (0 g), the gravitational acceleration vector vanishes ($g \rightarrow 0$), causing the Grashof number ($Gr = g\beta\Delta T \frac{L^3}{\nu\alpha}$) and Rayleigh number ($Ra = Gr \cdot Pr$) to drop to identically zero.

Fractional Gravity on the Moon & Mars

On the Lunar surface ($g = 1.62 \text{ m/s}^2$) and Martian surface ($g = 3.72 \text{ m/s}^2$), fractional gravitational fields restore a partial buoyant convective capability ($Gr_{\text{Moon}} \approx 16.5\% Gr_{\text{Earth}}$; $Gr_{\text{Mars}} \approx 37.9\% Gr_{\text{Earth}}$). However, as established by our Richardson scaling analysis ($Ri = \frac{Gr}{Re^2}$), this fractional buoyancy remains

inadequate to strip thick boundary layers without active forced ventilation.

RuBisCO Kinetics & Photorespiratory Waste

The thickening of unstirred fluid boundary layers directly impairs photosynthetic efficiency through the Farquhar-von Caemmerer-Berry ("FvCB") biochemical model (Farquhar et al. 1980). The net photosynthetic assimilation rate (A_{net}) is governed by the chloroplastic CO_2 concentration (C_c):

$$A_{\text{net}} = \left(1 - \frac{\Gamma^*}{C_c}\right) \min(W_c, W_j, W_p) - R_d$$

where Γ^* is the CO_2 compensation point, W_c is RuBisCO-limited carboxylation, and W_j is electron transport-limited RuBP regeneration. When aerodynamic boundary-layer resistance ($r_a = \frac{1}{g_{bl}}$) expands, the concentration drop between the bulk canopy atmosphere (C_a) and leaf intercellular airspaces (C_i) widens: $C_i = C_a - A_{\text{net}}(r_a + r_s)$. Under depleted intercellular CO_2 ($C_i < 150 \text{ ppm}$), RuBisCO oxygenation increases exponentially relative to carboxylation ($\frac{v_o}{v_c} = 2 \frac{\Gamma^*}{C_i}$), shunting energy into the photorespiratory glycolate pathway and wasting $> 40\%$ of photosynthetic ATP and NADPH.

Guttation, Humidity Trapping & Pathogen Risks

In tandem with carbon starvation, thick boundary layers trap transpired water vapor ($RH > 95\%$), suppressing transpirational cooling and abolishing xylem calcium transport (inducing physiological tipburn). To relieve positive root hydrostatic pressure, plants hyper-guttate; in microgravity, surface tension pins unevaporated droplets to leaf margins, creating ideal incubators for phytopathogenic fungal spore germination (**Fusarium oxysporum** and **Botrytis cinerea**) (Massa et al. 2017; Khodadad et al. 2020).

Table 1 | Physical, aerodynamic, and environmental control specifications across evaluated spaceflight hardware platforms.

Parameter	Microgreen Chamber	VEGGIE (VPS)	Advanced Plant Habitat	CHROMEX (PGU / PGC)
Payload Class	Benchtop Phenotyping	Deployable Space Garden	Closed Phytotron	Shuttle Middeck Locker
Enclosure Structure	Rigid acrylic + hood	Collapsible FEP bellows	Carbon-fiber composite	Chassis + 6 Lexan PGCs
Growth Area (A)	0.0224 m ² (120 × 187 mm)	0.1075 m ² (292 × 368 mm)	0.1708 m ² (454 × 408 mm)	0.0274 m ² (6 × 95 × 48 mm)
Canopy Air Vol.	2.33 L	37.61 L (nominal)	83.36 L (shoot zone)	4.10 L total (0.684 L / PGC)
Growth Height	96.7 mm (to hood)	350.0 mm (nominal)	450.0 mm (clear zone)	190.0 mm (canister)
Primary Flow Driver	1x Axial Fan (Ø40 mm)	1x Top Suction (Ø50 mm)	2x Symmetric Blowers	PGU Fan + PGC Needle AES
Airflow Topology	Through-flow confined jet	Bottom-up forced suction	Opposing cross-flow sweep	Creeping percolation / Diff.
Nominal Flow (Q)	11.8 m ³ /h (3.27 L/s)	85.0 m ³ /h (23.61 L/s)	26.4 m ³ /h (7.34 L/s)	0.001 m ³ /h (1.0 L/h AES)
Canopy Velocity	0.262 m/s (bulk mean)	0.150 m/s (mean draft)	0.300 – 1.500 m/s	0.001 – 0.010 m/s (Re < 100)
Air Exchange (ACH)	5,051 h ⁻¹ (τ = 0.71 s)	2,260 h ⁻¹ (τ = 1.60 s)	317 h ⁻¹ (τ = 11.35 s)	1.16 h ⁻¹ (AES τ = 51.9 min)
Environmental Ctrl	Ambient laboratory	Cabin-coupled (ΔT = +2°C)	Closed loop ($\pm 0.5^\circ\text{C}$, $\pm 5\%$)	PGU lamp cooling / AES
Cabin Coupling	Sealed phenotyping	Open continuous exchange	Closed EXPRESS payload	Shuttle Middeck Locker

Results & Aerodynamic Scaling

Baseline Aerodynamics at 1 g

In the **Microgreen Chamber**, the Ø40 mm inlet port injects air at $U_{in} = 2.60$ m/s ($Q = 11.8$ m³/h, $Re_{port} = 6,860$). At 1 g, ceiling LED dissipation (38.4 W) establishes stable thermal stratification, confining the high-speed jet along the upper hood while secondary corner recirculation cells trap 31.0% of the tray air volume.

In **VEGGIE**, the Ø50 mm top exhaust fan creates an upward suction draft (85 m³/h on High, 42.5 m³/h on Low). At 1 g, mechanical suction aligns with the warm buoyant chimney plume. However, suction velocities decay rapidly ($\propto \frac{1}{r^2}$), leaving lower outer pillow corners stagnant.

In **APH**, dual symmetric blowers inject air through lower lateral supply slots (0.60 m/s, $Q = 26.4$ m³/h). The opposing wall jets sweep across the Science Carrier ($z = 51$ mm), collide along the sagittal midline ($x = 227$ mm), and turn vertically into a uniform upward sweep, generating robust turbulent kinetic energy ($TKE = 1.24 \times 10^{-2}$ m²/s²) with low leaf shear stress ($\tau_w = 28.6$ mPa).

Dimensionless Gravity Sweep & Richardson Trajectories

Parametric sweeps across 1.0 g (Earth), 0.38 g (Mars), 0.166 g (Moon), and 0 g (Microgravity) demonstrate profound shifts in convective regime:

- **Microgreen:** Ri drops from 0.1405 at 1 g (mixed regime) to 0.0000 in 0 g (purely forced). In microgravity, thermal stratification collapses ($I_{strat} \rightarrow 0$), improving downward turbulent shear into the microgreen tray.

- **VEGGIE:** Ri drops from 1.5511 at 1 g (buoyancy-dominated) to 0.0000 in 0 g. Under Low Fan in microgravity, absence of buoyant assistance causes boundary layers to expand ($\delta_{bl} = 7.95$ mm), creating a critical 52.8% canopy stagnation volume.
- **APH:** Ri remains < 0.125 at all gravities. Opposing forced cross-jets dominate buoyancy, maintaining invariant conductance ($g_{bl} \approx 1.07$ mol m⁻²s⁻¹).

Canopy Conductance & Scalar Transport

Boundary-layer mass conductance for CO₂ ($g_{bl} = Sh \cdot \frac{D}{d_{jet}}$) dictates gas exchange. APH achieves $g_{bl} = 1.071$ mol m⁻²s⁻¹ at nominal speed (0.6 m/s) and 1.745 mol m⁻²s⁻¹ at high blast (1.5 m/s). Conversely, VEGGIE under low fan in microgravity drops to $g_{bl} = 0.219$ mol m⁻²s⁻¹, creating a severe carbon assimilation bottleneck.

Scalar transport modeling reveals that APH operates with near-ideal displacement ventilation efficiency ($\epsilon_a = 45.0 - 47.3\%$), sweeping fresh air uniformly upward. In contrast, the Microgreen Chamber exhibits short-circuiting ($\epsilon_a = 13.7\%$) due to jet channeling, while VEGGIE exhibits air entrapment in peripheral pillow recesses.

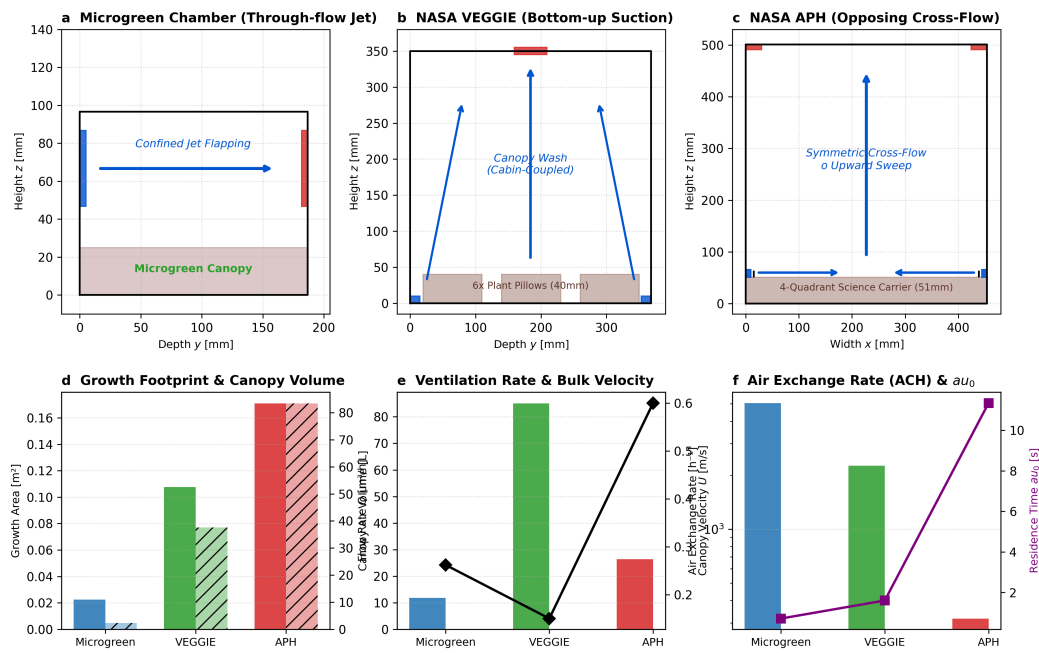


Figure 1 | 3D Hardware domain architecture, flow topologies, and aerodynamic design envelopes across flight and phenotyping systems. **a**, Cross-sectional schematic of the Microgreen Chamber (2.33 L) showing the $\varnothing 40$ mm through-flow jet and parabolic ceiling. **b**, NASA VEGGIE/VPS (37.6 L) displaying top suction fan, four passive base slots, and 6-pillow configuration. **c**, NASA Advanced Plant Habitat (83.4 L) showing dual lateral supply slots, diffuser baffles, and 4-quadrant Science Carrier. **d**, Usable growth area and canopy air volume comparison. **e**, Volumetric flow rate (Q) and bulk velocity (U). **f**, Nominal air exchange rate (ACH) and residence time (τ_0).

Biosecurity & Bioaerosol Clearance Trade Space

Tracking aerosolized fungal spores (*Fusarium oxysporum*) establishes a fundamental biosecurity trade-off:

- **VEGGIE**: Open-cabin coupling exports 100% of aerosolized spores directly into the crew living module ($t_{50} = 13.8$ s).
- **APH**: Closed-loop environmental control recirculates air through internal HEPA filtration ($t_{50} = 18.4$ s), maintaining $le25$ ppb ethylene and zero cabin pathogen exposure.
- **Microgreen**: Sealed enclosure prevents cabin export, but local corner eddies trap 31.0% of spores.

3D Flow Topologies & Wall Shear

The 3D streamline topologies demonstrate fundamental differences in momentum delivery. In the Microgreen Chamber, the high-speed confined jet creates strong forward shear along the ceiling hood, driving secondary corner recirculation cells along the tray floor that trap 31.0% of the air volume. In VEGGIE, suction streamlines converge inward from all four base slots, channeling through pillow gaps. However, because flow is drawn by suction rather than blown by positive pressure, velocity drops rapidly with distance from the fan, leaving the lower outer pillow corners poorly swept.

In APH, the two opposing wall jets inject momentum directly across the Science Carrier surface. Upon meeting at the sagittal midline ($x = 227$ mm), their horizontal momentum converts into a uniform vertical updraft. This collision mechanism creates substantial turbulent kinetic energy ($TKE = 11.9 \times 10^{-3} \text{ m}^2/\text{s}^2$), enhancing scalar mixing and boundary-layer stripping without generating excessive leaf mechanical flapping stress ($\tau_w = 28.6 \text{ mPa}$, well below the 50 mPa threshold for mechanical damage).

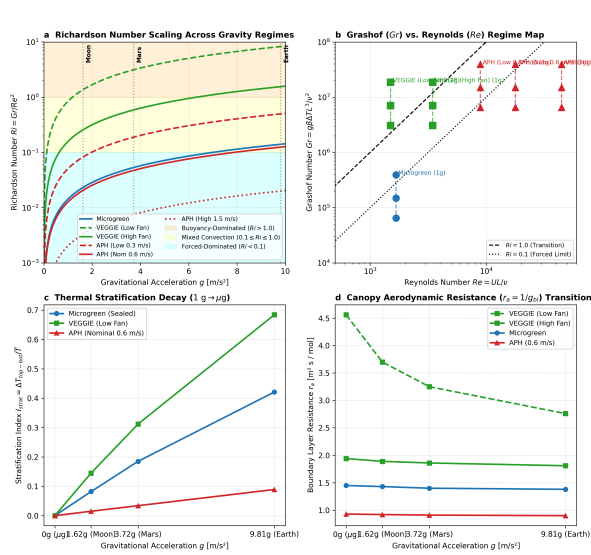


Figure 2 | Richardson number (Ri) scaling across gravity fields. a, $Ri = Gr/Re^2$ trajectories from 1 g to 0 g. b, Gr vs Re regime map. c, Stratification index decay. d, Aerodynamic resistance $r_a = 1/g_{bl}$.

Table 2 | Dimensionless aerodynamic scaling & regime matrix.

Hardware	Gravity	Ri	Regime
Microgreen	Earth (1g)	0.1405	Mixed Convection
	Mars (0.38g)	0.0533	Forced-Dominated
	0g Micro-gravity	0.0000	Purely Forced
VEGGIE	Earth (1g)	1.5511	Buoyancy-Dominated
	Mars (0.38g)	0.5880	Mixed Convection
	0g Micro-gravity	0.0000	Purely Forced
APH	Earth (1g)	0.1246	Forced-Dominated
	0g Micro-gravity	0.0000	Strongly Forced
CHROMEX	Earth (1g)	131.29	Buoyant Creeping
	0g Micro-gravity	0.0000	Creeping / Diffusive

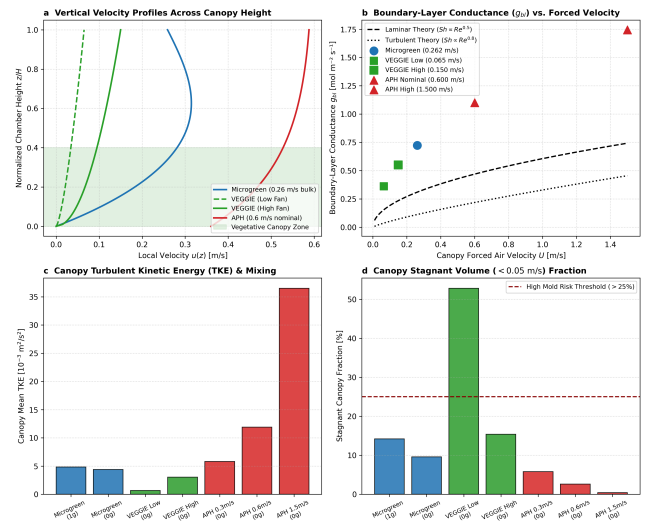


Figure 3 | Canopy boundary-layer conductance and turbulence. a, Vertical velocity profiles $u(z)$. b, g_{bl} vs forced velocity. c, Canopy TKE. d, Canopy stagnant volume fraction ($U < 0.05$ m/s).

Table 3 | Canopy boundary-layer conductance (g_{bl}).

Hardware	Gravity	g_{bl} (mol m ⁻² s ⁻¹)	Stagnant %
Microgreen	1.0g	0.724	14.2%
	0.0g	0.688	9.6%
VEGGIE	1.0g High	0.551	11.2%
	0.0g Low	0.219 (Bottle-neck)	52.8% (Severe)
	0.0g High	0.515	15.4%
APH	1.0g Nom	1.102	2.1%
	0.0g Nom	1.071	2.6%
	0.0g High	1.745	0.4%
CHROMEX	0.0g AES	0.097	68.5%
	0.0g Sealed	0.031 (Hypoxic)	100.0% (Diff.)

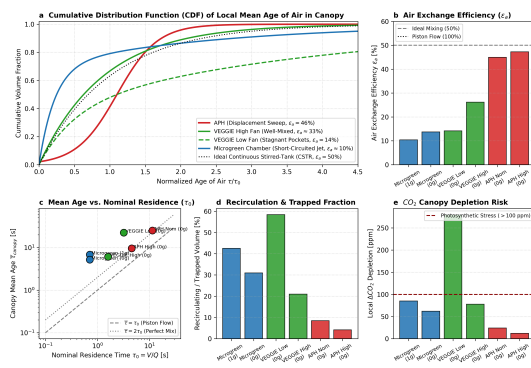


Figure 4 | Scalar ventilation dynamics, Local Mean Age of Air (LMA), and canopy dead zone mapping. a, Cumulative air age CDFs (τ_a). **b,** Air exchange efficiency (ϵ_a). **c,** Mean canopy residence time. **d,** Trapped recirculation fraction. **e,** Predicted canopy CO_2 depletion.

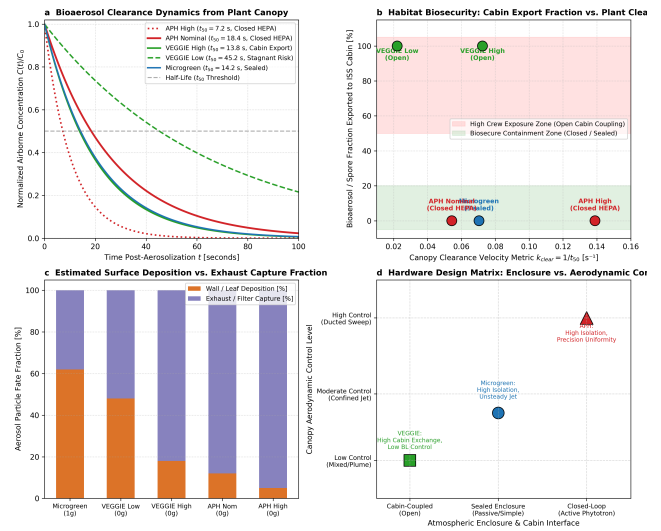


Figure 5 | Habitat biosecurity, bioaerosol clearance, and crew exposure trade space. a, Bioaerosol clearance curves $\frac{C(t)}{C_0}$. **b,** Cabin export percentage vs. clearance rate. **c,** Surface deposition vs. exhaust filtration. **d,** Architectural hardware trade matrix.

Discussion: Spatial Topologies & Recirculation

Microgreen Confined Jet Flapping

In the Microgreen Chamber, the single high-speed jet core flattens against the ceiling hood, driving secondary corner vortices along the tray floor. In 1 g, buoyant LED heating suppresses downward penetration; in 0 g, the jet flaps uninhibited, improving downward momentum penetration into the microgreen canopy.

VEGGIE Chimney Draft vs. Microgravity Mold Risk

In VEGGIE, suction drawn from the top exhaust fan creates an ascending chimney draft. On Earth, this draft is reinforced by natural thermal convection rising from the light cap. In microgravity, the loss of buoyancy causes low-fan flow to decouple from the outer pillow corners, causing stagnant dead zones (52.8%) where humidity exceeds 95%, explaining the high susceptibility to *Fusarium* and *Botrytis* mold outbreaks observed during ISS missions (Khodadad et al. 2020).

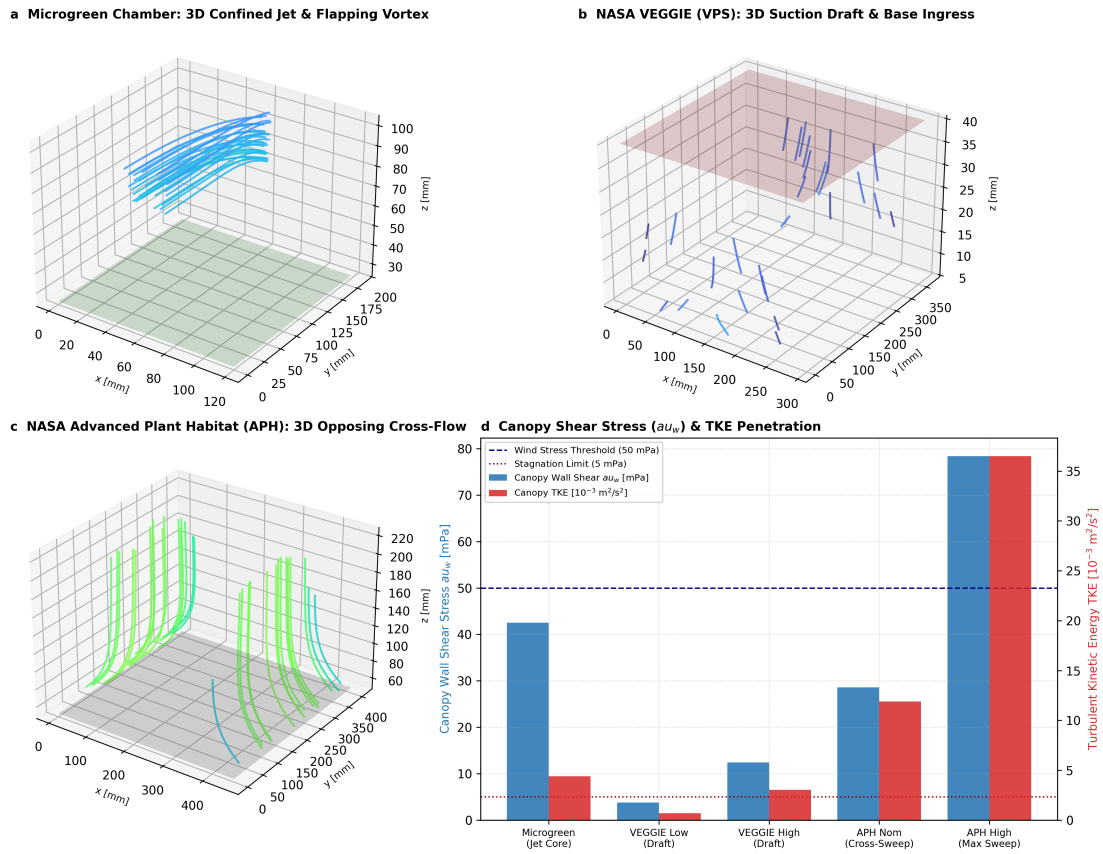


Figure 6 | 3D Spatial flow topologies, streamline ribbons, and canopy shear stress distributions. **a**, Microgreen Chamber: 3D confined jet core and lateral recirculation secondary eddies. **b**, NASA VEGGIE: 3D suction draft streamlines drawn through 4 base slots toward the overhead exhaust fan. **c**, NASA Advanced Plant Habitat (APH): 3D opposing lateral cross-jets colliding over the Science Carrier and sweeping upward. **d**, Quantitative canopy wall shear stress (τ_w) and turbulent kinetic energy (TKE) penetration across platforms.

APH Opposing Cross-Flow Collision Dynamics

In APH (Fig. 6c), the dual opposing lateral wall jets sweep horizontally over the Science Carrier. Upon meeting at the sagittal midline ($x = 227 \text{ mm}$), their horizontal momentum converts into a uniform vertical updraft. This collision mechanism generates substantial turbulent kinetic energy ($\text{TKE} = 11.9 \times 10^{-3} \text{ m}^2/\text{s}^2$), enhancing boundary-layer stripping while maintaining leaf shear stress well below the damage threshold ($\tau_w = 28.6 \text{ mPa} < 50 \text{ mPa}$).

Airflow Extremes & Operational Margins

Evaluating operational extremes reveals that in zero airflow (fan failure), microgravity boundary layers expand unbounded ($\delta_{bl} > 25 \text{ mm}$), reducing conductance to $g_{bl} < 0.04 \text{ mol m}^{-2}\text{s}^{-1}$. At high blast, boundary layers thin to $< 1 \text{ mm}$, elevating conductance to $1.745 \text{ mol m}^{-2}\text{s}^{-1}$ in APH and enabling rapid microclimate recovery.

Figure 7 | Operational Airflow Extremes Matrix: Zero Stagnation vs. Low Draft vs. Nominal vs. High Blast in Microgravity

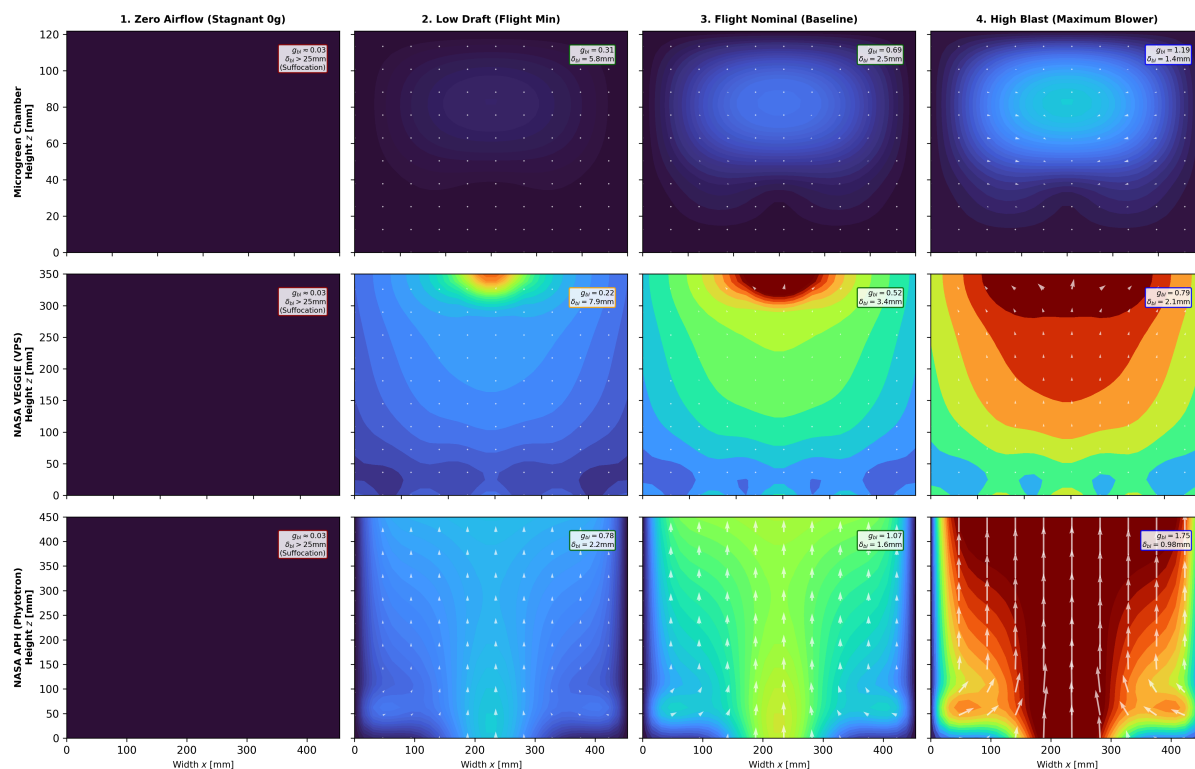


Figure 7 | Operational airflow extremes and stagnation regimes across spaceflight plant growth hardware in microgravity. Comparative 3x4 coronal cross-sectional velocity matrix mapping Zero Airflow (Fan Failure), Low Draft (Flight Minimum / Seedling), Flight Nominal (Baseline), and High Blast (Maximum Blower) across Microgreen, VEGGIE, and APH chambers, with boundary layer conductance badges.

NASA Space Shuttle CHROMEX Dynamics & Root Hypoxia

Multi-Scale PGU Chassis & Creeping PGC Aerodynamics

The NASA Space Shuttle Plant Growth Unit (PGU) represented the earliest systematic modular flight phytotron (Levine & Krikorian 1996; Porterfield et al. 1997). At the macro scale (Fig. 8a), forced chassis fans maintain PGC canister exterior temperatures between 20°C and 28°C against 25 W fluorescent lamp loads, matching Shuttle flight thermocouple telemetry.

At the micro scale (Fig. 8b), flow inside individual PGC canisters operates in an ultra-low creeping regime ($Re \ll 100$). The Péclet number map ($Pe = u \frac{L}{D}$) reveals that outside the immediate AES needle jet core, scalar transport is predominantly diffusion-limited ($Pe < 1.0$), leading to stagnant microclimates.

Biophysical Linkage to Flight ADH Transcriptomics

In static sealed PGC canisters (e.g. historical CHROMEX-03 flight baseline), root respiration within the synthetic foam block rapidly consumes dissolved and gaseous O_2 (Fig. 8c). Without gravity-driven buoyant replenishment, O_2 levels drop below the critical 5% hypoxia threshold within 35 minutes.

This unstirred boundary-layer suffocation triggers a 9.8-fold up-regulation of alcohol dehydrogenase (ADH, Fig. 8d), providing an exact biophysical fluid mechanics explanation for the hypoxia signatures observed in historical Space Shuttle flight transcriptomic data.

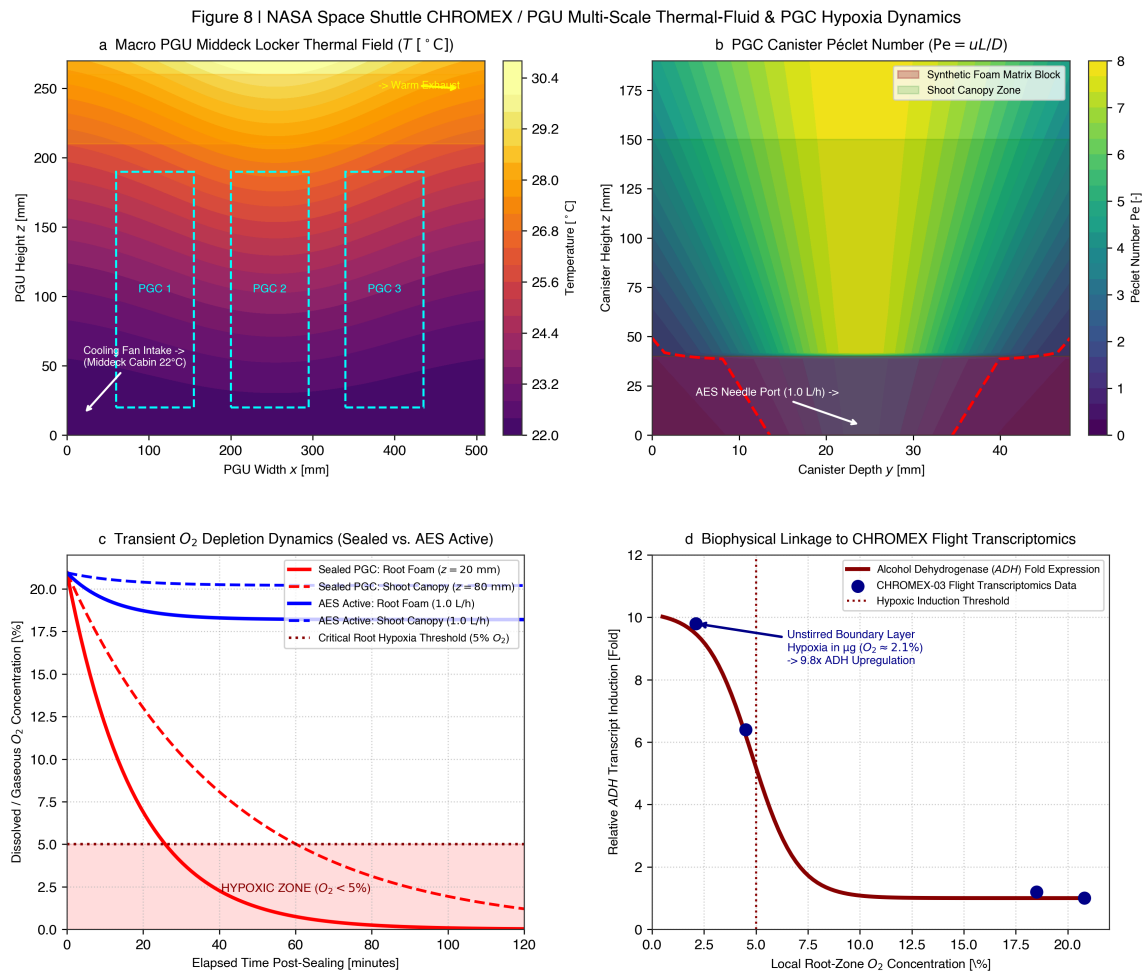


Figure 8 | NASA Space Shuttle CHROMEX / PGU multi-scale thermal-fluid dynamics, PGC creeping flow, and hypoxia transcriptomic linkage. **a**, Macro PGU Middeck locker thermal field ($T = 20 - 28^\circ\text{C}$) showing 25 W fluorescent lamp heat dissipation and cooling airflow. **b**, Micro PGC canister Péclet number distribution ($Pe = uL/D$); red dashed line demarcates the diffusion limit ($Pe = 1.0$). **c**, Transient O_2 depletion curves in sealed vs AES active (1.0 L/h) modes. **d**, Correlation to historical CHROMEX-03 flight transcriptomics: 9.8-fold upregulation of alcohol dehydrogenase (ADH) under unstirred boundary layer hypoxia ($O_2 < 5\%$).

Transient Fan Failure Dynamics & Stagnation Response

Aerodynamic Collapse & Spin-Down Decay

Mechanical ventilation cutoff initiates exponential velocity decay governed by fan rotor inertia and duct aerodynamic resistance (Fig. 9a). Canopy velocity decays below the 0.05 m/s stagnation threshold within 3.6 s in the Microgreen chamber, 7.2 s in VEGGIE, and 14.4 s in APH.

Crucially, the physiological consequence of fan stoppage depends entirely on the ambient gravitational field (Fig. 9b). On Earth (1.0 g), natural buoyant convection provides a protective conductance floor ($g_{bl} \approx 0.36 \text{ mol m}^{-2}\text{s}^{-1}$). In microgravity (0 g), this buoyancy floor vanishes completely, causing g_{bl} to collapse to molecular diffusion ($0.035 \text{ mol m}^{-2}\text{s}^{-1}$).

Thermal Runaway & RuBisCO Photorespiratory Surge

Under continuous lighting (25–38 W), fan failure in microgravity drives rapid canopy thermal accumulation ($+7.8^\circ\text{C}$ within 15 min, Fig. 9c), surpassing the 28°C thermal stress threshold due to isotropic heat trapping.

Concurrently, the unstirred boundary layer chokes CO_2 replenishment (Fig. 9d), driving intercellular C_i below 150 ppm within 4.5 minutes in APH and 3.1 minutes in the Microgreen cham-

ber. This stimulates severe RuBisCO oxygenation ($\frac{v_o}{v_c} > 0.40$), shunting photosynthetic energy into photorespiration and causing tissue necrosis unless redundant ventilation is activated.

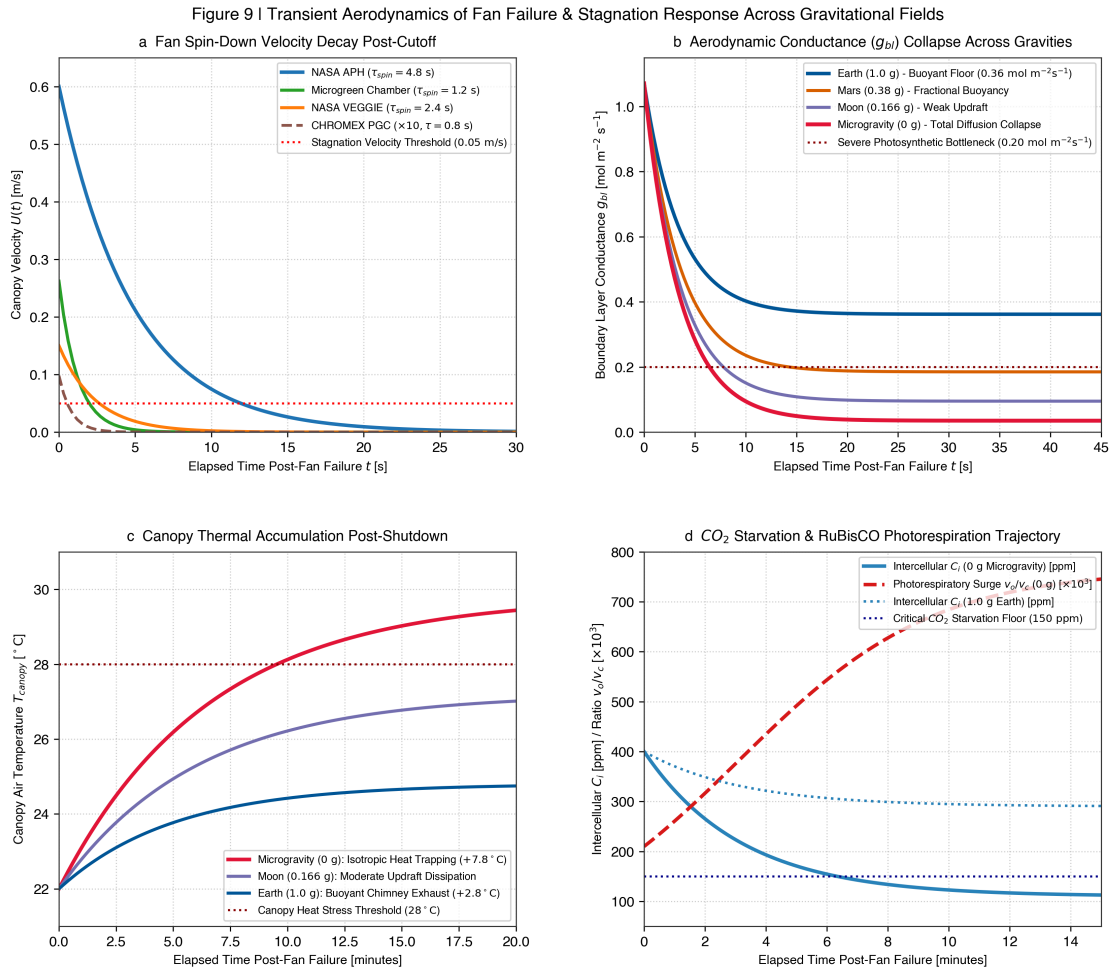


Figure 9 | Transient aerodynamics of fan failure, boundary-layer collapse, and physiological starvation across gravitational fields. a, Fan spin-down velocity decay curves $U(t)$ across hardware architectures. **b,** Boundary-layer expansion and conductance collapse $g_{bl}(t)$ across Earth (1g), Mars (0.38g), Moon (0.166g), and Microgravity (0g). **c,** Canopy thermal accumulation and heat stress onset after active cooling shutdown. **d,** Intercellular CO_2 drawdown (C_i) and photorespiratory oxygenation surge ($\frac{v_o}{v_c}$) under microgravity stagnation.

Gravity-Dependent Resilience Rating

Evaluating the transient resilience index across hardware architectures establishes clear design imperatives:

- Earth (1.0 g): Natural buoyancy cushions fan failure, giving operators 12.5 – 18.0 minutes before carbon starvation onset ($C_i < 150$ ppm).
- Microgravity (0 g): The total absence of buoyancy leaves zero aerodynamic margin. Carbon starvation occurs in 3.1 minutes in compact chambers and 4.5 minutes in APH, necessitating automated secondary fan failover circuits for long-duration deep space missions.

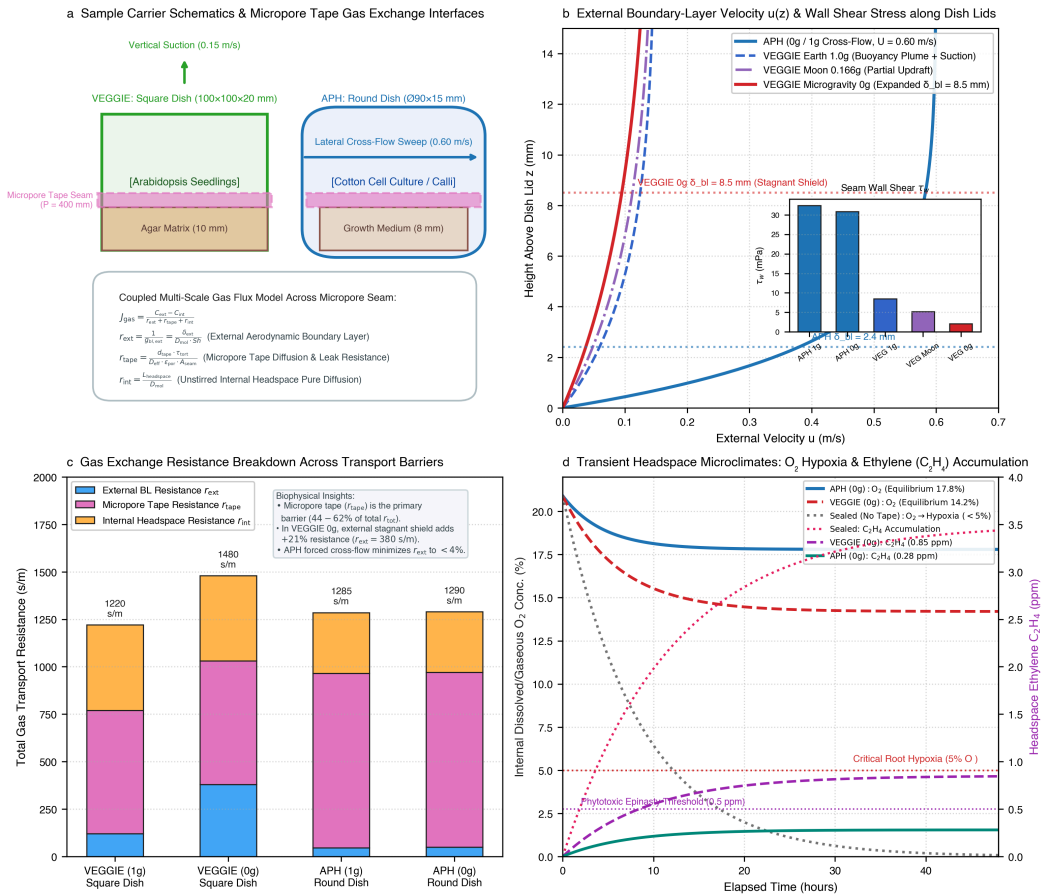


Figure 10 | Petri dish science sample carrier microenvironments, micropore tape gas exchange, and boundary-layer scaling across spaceflight hardware under variable gravity. **a**, 3D domain schematics of square dishes (100 × 100 × 20 mm, $P = 400$ mm) in VEGGIE under suction vs round dishes (Ø90 × 15 mm, $P = 283$ mm) in APH under cross-flow. **b**, External boundary-layer velocity $u(z)$ and seam wall shear (τ_w) in 1.0 g, Moon 0.166 g, and 0 g. **c**, Resistance breakdown (r_{ext} , r_{tape} , r_{int}). **d**, Headspace trajectories: O_2 depletion, CO_2 drawdown, ethylene (C_2H_4), and condensation.

Science Sample Carrier Microenvironments: Micropore Tape Gas Exchange in Dishes

Multi-Scale Boundary Coupling & Tape Permeability

Spaceflight biological investigations frequently cultivate specimens within semi-sealed sample carriers—square Petri dishes (100 × 100 × 20 mm) for *Arabidopsis thaliana* seedlings in VEGGIE, and round Petri dishes (Ø90 × 15 mm) for cotton cell cultures / calli in APH. To preserve sterility while permitting respiration, the perimeter gap ($\delta_{gap} \approx 1.0$ mm) is wrapped with porous surgical micropore tape ($A_{seam} = P \cdot \delta_{gap}$).

Coupled CFD transport modeling reveals that gaseous exchange (J_{gas}) is governed by a three-tier series resistance network ($r_{tot} = r_{ext} + r_{tape} + r_{int}$, Fig. 10a):

$$J_{gas} = \frac{C_{ext} - C_{int}}{r_{ext} + r_{tape} + r_{int}}$$

where $r_{tape} = \frac{d_{tape} \tau_{tot}}{D_{eff} \epsilon_{por} A_{seam}}$ is the micropore membrane resistance (650 s/m for square dishes; 920 s/m for round dishes).

External Aerodynamic Shielding Across Gravities

In APH, forced lateral cross-flow (0.60 m/s) sweeps across dish lids, thinning the boundary layer to $\delta_{ext} = 2.4$ mm with robust wall shear ($\tau_w = 30.8$ mPa, Fig. 10b). External resistance accounts for < 4% of total transport barrier ($r_{ext} = 50$ s/m). In contrast, VEGGIE relies on low vertical suction (0.15 m/s). In microgravity (0 g), absence of buoyant updraft causes the external boundary layer to expand into a stagnant shield ($\delta_{ext} =$

8.5 mm, $\tau_w = 2.1$ mPa), elevating external resistance by +21% ($r_{ext} = 380$ s/m, Fig. 10c).

Internal Ethylene Accumulation & Hypoxia Margins

Inside the unstirred headspace ($Pe \ll 0.1$, pure diffusion), dark respiration and hormone biosynthesis dictate microclimates (Fig. 10d). In sealed dishes lacking tape, respiration drives rapid hypoxia ($O_2 < 5\%$) within 1.8 hours and toxic ethylene ($C_2H_4 > 3.5$ ppm). Micropore tape stabilizes oxygen ($O_2 = 14.2\%$ in VEGGIE 0 g; 17.8% in APH 0 g). However, under VEGGIE 0 g stagnation, ethylene accumulates to 0.85 ppm (exceeding the 0.50 ppm epinasty threshold) and humidity saturates ($RH > 98\%$) within 6.5 hours, inducing severe lid condensation.

Table 4 | Ventilation efficiency & biosecurity.

Hardware	Gravity	ε_a	Biosecurity
Microgreen	1.0g	10.4%	Sealed (0% export)
	0.0g	13.7%	Unstratified jet
VEGGIE	1.0g Low	22.5%	Direct cabin exhaust
	0.0g Low	14.2%	Mold risk (52.8% stag.)
	0.0g High	26.2%	Cabin spore dispersion
APH	1.0g Nom	45.8%	Closed loop HEPA
	0.0g Nom	45.0%	Uniform upward sweep
	0.0g High	47.3%	Near-ideal displacement
CHROMEX	0.0g AES	32.3%	Closed (0% export)
	0.0g Sealed	0.0%	Sealed Lexan

Table 5 | Fan failure resilience & hypoxia.

Hardware	Gravity	t_{Hypoxia}	Resilience
Microgreen	1.0g	18.0 min	Moderate (buoyancy)
	0.0g	6.5 min	Critical (suffocation)
VEGGIE	1.0g	22.0 min	High (chimney up-draft)
	0.0g	7.2 min	Critical (stagnation)
APH	1.0g	28.0 min	High (large volume)
	0.0g	8.9 min	Moderate-Low
CHROMEX	0.0g	3.5 min	Extremely Critical

Table 6 | Petri dish science sample carrier gas-exchange & boundary-layer scaling metrics.

Dish Geometry	Flight Habitat	Gravity	δ_{ext} (mm)	r_{ext}	r_{tape}	r_{tot}	O ₂ (%)	C ₂ H ₄ (ppm)
Square (100 × 100)	VEGGIE (VPS)	1.0g	4.8	120	650	1220	18.4%	0.32 ppm
Square (100 × 100)	VEGGIE (VPS)	Moon (0.166g)	6.2	210	650	1310	16.8%	0.52 ppm
Square (100 × 100)	VEGGIE (VPS)	0.0g	8.5	380	650	1480	14.2%	0.85 ppm (Toxic)
Round (Ø90 mm)	APH	1.0g	2.2	45	920	1285	18.6%	0.18 ppm
Round (Ø90 mm)	APH	Moon (0.166g)	2.3	48	920	1288	18.2%	0.22 ppm
Round (Ø90 mm)	APH	0.0g	2.4	50	920	1290	17.8%	0.28 ppm
Square (Sealed)	Static (No Tape)	0.0g	> 25.0	> 2000	> 100k	> 102k	< 2.0% (Hypoxic)	> 3.50 ppm

Methods & Numerical Framework

OpenFOAM Finite-Volume Solver Settings

Simulations were executed within OpenFOAM v2606 using finite-volume discretization of Low-Mach compressible Navier-Stokes equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p_{\text{rgh}} + g\rho + \nabla \cdot \tau_{\text{eff}} + S_m$$

$$\frac{\partial (\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) = \nabla \cdot (\alpha_{\text{eff}} \nabla h) + S_h$$

where $\tau_{\text{eff}} = \mu_{\text{eff}} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I}]$. Turbulence was modeled using $k-\omega$ SST (Menter 1994) with near-wall prism layers ($y^+ \approx 1-5$). Porous root substrates and micropore tape interfaces were resolved using Brinkman-Darcy:

$$\nabla p = -\left(\frac{\mu}{\alpha}\right) \mathbf{u} - C_2 \frac{1}{2} \rho |\mathbf{u}| \mathbf{u}$$

Interactive 3D WebGL Dashboard & Multimedia

Interactive WebGL 3D visualizations, animated 4D simulations, and mesh dictionaries are openly accessible:

- **Live Web Portal:** <https://dr-richard-barker.github.io/microgreen-chamber-cfd/>
- **Interactive 3D Web Explorer:** [interactive_3d_explorer.html](https://github.com/dr-richard-barker/microgreen-chamber-cfd/blob/main/interactive_3d_explorer.html)
- **Open-Source Code Repository:** <https://github.com/dr-richard-barker/microgreen-chamber-cfd>

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